

ANIMESH RAJ

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EDUCATION

Indian Institute of Technology, Kharagpur

B.Tech (Hons.) in Instrumentation Engineering | CGPA: 8.54/10

Kharagpur, India

Aug 2023 – Apr 2027

Minor: Artificial Intelligence & Computer Science Micro-Specialization: High-Performance Computing

Delhi Model Public School

Class XII AISSCE (Science Stream) | Aggregate: 94.2%

Patna, India

Completed 2023

EXPERIENCE

Stealth AI Startup

Remote

Software Engineer Intern (R&D) | LangGraph, WebRTC, LiveKit, Vapi

Oct 2025 – Nov 2025

- Co-developed a proprietary Voice AI observability platform MVP, now operating in stealth with unicorn-status enterprise clients.
- Engineered a graph-based engine converting unstructured prompts into test suites, reducing conversational AI QA cycles by **65%**.
- Developed comprehensive metrics suite tracking latency, tone, and logic flows to automate stress-testing for generative voice models.

Convosight

Remote

Software Engineering Intern | React, Supabase, NLP

Aug 2025 – Oct 2025

- Migrated legacy workflows to a high-performance React/Supabase architecture, cutting infra costs by **38%** and load times by **50%**.
- Built an ETL pipeline processing **10k+ daily posts** with **92%** sentiment accuracy via NLP models for cross-market trend detection.
- Automated executive insight summarization via fine-tuned GPT systems, reducing manual content creation overhead by over **70%**.

Aristhrottle

Remote

AI/ML Engineer | LangGraph, PostgreSQL, FastAPI

June 2024 – July 2024

- Built a production AI pipeline for automated story generation, ensuring visual coherence via Google Imagen3 API and PostgreSQL.
- Designed a stochastic prompt selector with temperature scaling, increasing the creative output diversity by **25%** in A/B tests.
- Integrated **LangFuse SDK** to trace LangGraph's Stategraph executions, enabling granular monitoring of token usage and latency.

PROJECTS

Aksha Drishti: Agentic Vision-Language System for Remote Sensing Imagery

Nov 2025 – Dec 2025

Gold Medal Winner, ISRO Inter IIT Tech Meet 14.0 | vLLM, Qwen3-VL, SAM3

- Achieved **2-3x** latency reduction and **95%** GPU utilization via vLLM service optimized with KV caching and concurrent batching.
- Outperformed **GPT-4o** on VRSBench by developing a DPO-aligned **Qwen3-VL-8B** pipeline, reaching **87% VQA accuracy**.
- Achieved **mIoU** score of **0.73** in grounding via FastAPI **SAM3 Agent** service with pyramidal tiling, cross-verified by VLM logic.

Financial Trading Intelligence Agent | CrewAI, Plotly, SEC EDGAR

Sept 2025

- Engineered a **CrewAI** multi-agent system synthesizing real-time SEC EDGAR, News API and X data to identify market anomalies.
- Designed interactive **Plotly dashboards** to visualize intraday trading signals, market sentiment and historical insider transactions.
- Built robust web scraper processing thousands of data points daily with **99% uptime** for real-time intelligence using Crawl4ai.

RESEARCH & PUBLICATIONS

RAD-Agent: Voice-Driven Dataset Automation | Microsoft Presidio, Whisper

Mar 2025 – Present

- Built an agentic pipeline converting doctor's dictation into structured datasets, cutting annotation costs by **80%** with F-score > 0.80.
- Ensured **100% HIPAA compliance** via Microsoft Presidio for PHI removal and integrated Whisper for hands-free transcription.

COMPETITIONS

Budget Optimization | Data Analytics GC 2025 | Bronze Winning Team

Mar 2025

- Engineered Marketing Mix Model with Meta's Robyn library to optimize an **\$18.7M** yearly budget, projected **31.9%** revenue uplift.

Employee Selection Engine | Open Soft GC 2025 | Bronze Winning Team

Apr 2025

- Architected predictive analytics pipeline achieving **87%** accuracy in at-risk employee identification using **6** heterogenous HR datasets

TECHNICAL SKILLS

Languages & Core: Python, C++, JavaScript, R, SQL (PostgreSQL), HTML5/CSS3

AI & Agentic Systems: LangGraph, LangChain, PyTorch, vLLM, HuggingFace, OpenCV, YOLO, SAM, Whisper

Web & Backend: React.js, FastAPI, Node.js, PostgreSQL, Supabase, Pinecone, FAISS, MongoDB, WebRTC

Tools & Infrastructure: Docker, AWS S3, Modal, Git/GitHub, Linux (HPC Environments), LangFuse, LangSmith

RELEVANT COURSEWORK

Computer Science / AI: Data Structures, Algorithms, Introduction to AI/ML, DBMS, High-Performance Computing (HPC)

Systems / Mathematics: Signals & Systems, Control Systems, Digital Signal Processing (DSP), Probability, Linear Algebra